Jinpeng Chen

Objective: AI Scientist/Engineer

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SUMMARY

- I am a third-year Ph.D. candidate in the Department of Computer Science at the City University of Hong Kong, expecting to graduate between June and October 2025.
- During my studies, my research has primarily focused on continual learning and image segmentation. I have published five papers, three of which I am the first author. Additionally, I have two more papers under review, in which I am the first or co-first author.
- I am currently a research intern at Huawei Hong Kong Research Center, where I conduct research on multimodal large language models (MLLMs). This includes developing AI agents based on MLLMs and exploring continual learning techniques for MLLMs.
- I am seeking a research or engineering position related to AI, starting in the fall of 2025.

EDUCATION

Ph.D. | City University of Hong Kong **Sep 2021 - Present**

- Major: Computer Science
- Supervisor: Prof. Horace Ip and Prof. Sam Kwong (IEEE Fellow, NAI Fellow)
- Research Focus: Continual Learning, MLLMs, Image Segmentation
- GPA: 3.90/4.3

B.Eng. | Beijing University of Posts and Telecommunications **Sep 2017 - Jun 2021**

- Major: Electronic Information Science and Technology
- GPA: 3.63/4.0

EMPLOYMENT

Research Intern | Huawei Hong Kong Research Center **May 2024 - Present**

- Mentor: Dr. Yuzhi Zhao
- Developing AI agents for smartphones utilizing MLLMs, with a focus on decision-making and self-reflection strategies, as well as implementing Retrieval-Augmented Generation (RAG) methods.
- Conducting research on continual learning techniques for MLLMs to ensure the retention of pre-training knowledge and generalization capabilities during fine-tuning with new data.

Research Intern | Tsinghua University **Aug 2019 - Jun 2021**

- Researched the use of Photoplethysmography (PPG) signals for user authentication, aiming to address the limitations of traditional biometric solutions in defending against non-liveness attacks.

PUBLICATIONS

[1] **J. Chen**, R. Cong, Y. Luo, H. Ip, and S. Kwong, "Saving 100x storage: Prototype replay for reconstructing training sample distribution in class-incremental semantic segmentation," *NeurIPS 2023*.

[2] **J. Chen**, R. Cong, Y. Luo, H. Ip, and S. Kwong, "Strike a balance in continual panoptic segmentation," *ECCV 2024*.

[3] **J. Chen**, R. Cong, H. Ip, and S. Kwong, "KepSalinst: Using peripheral points to delineate salient instances," *IEEE Transactions on Cybernetics*, 2023.

[4] **J. Chen**, R. Cong, Y. Luo, H. Ip, and S. Kwong, "Replay without saving: Prototype derivation and distribution rebalance for class-incremental semantic segmentation," under review at *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.

[5] Y. Luo*, **J. Chen***, R. Cong, H. Ip, and S. Kwong, "Trace back and go ahead: Completing partial annotation for continual semantic segmentation," under review at *NeurIPS 2024*. (*Equal Contribution)

[6] R. Cong, Y. Guan, **J. Chen**, W. Zhang, Y. Zhao, and S. Kwong, "SDDNet: Style-guided dual-layer disentanglement network for shadow detection," *ACM MM 2023*.

[7] R. Cong, H. Xiong, **J. Chen**, W. Zhang, Q. Huang, and Y. Zhao, "Query-guided prototype evolution network for few-shot segmentation," *IEEE Transactions on Multimedia*, 2024.

PROJECTS

Smartphone AI Agent Based on MLLMs **May 2024 - Present**
(During the Internship at Huawei Hong Kong Research Center)

- **Widget Perception and Grounding:** Developing precise widget perception and grounding solutions based on smartphone screenshots and layout files. This involves integrating techniques such as OCR, object detection, and grid line annotations to accurately identify and locate UI elements.
- **Prompt Optimization:** Enhancing the AI agent's accuracy in planning, decision-making, and self-reflection by employing various prompting techniques, including chain-of-thought (CoT) and few-shot learning strategies.
- **Exploration of RAG Technology:** Exploring the application of RAG technology in the smartphone AI agent. This includes developing a retrieval method based on generalized page descriptions and creating a framework to summarize annotated operation sequences into user documentation for the AI agent's reference.

Fingertip PPG Signal-Based User Authentication Algorithm and System Prototype **Jan 2021 - Jun 2021**
(During the Internship at Tsinghua University)

- **Device Design and Assembly:** Designed and assembled a fingertip PPG signal acquisition device using Arduino.
- **Data Collection Experiment:** Organized and conducted experiments to collect PPG signals from participants.
- **User Authentication Algorithm:** Designed and implemented a user authentication algorithm for fingertip PPG signals using a temporal convolutional network and a triplet loss function.
- **System Prototype Implementation:** Created a user authentication application with a graphical user interface, compatible with both Windows and Raspberry Pi platforms. This resulted in a complete system prototype ready for practical deployment, effectively preventing non-liveness attacks.

HONORS AND AWARDS

- **Outstanding Reviewer**, IEEE Journal of Oceanic Engineering 2023
- **Outstanding Presentation Award**, The 3rd Graduate Forum of the China Society of Image and Graphics 2023
- **2nd Class Scholarship**, Beijing University of Posts and Telecommunications 2020
- **Academic Progress Award**, Beijing University of Posts and Telecommunications 2019
- **Enterprise Scholarship**, Beijing University of Posts and Telecommunications 2019
- **3rd Class Scholarship**, Beijing University of Posts and Telecommunications 2018

SKILLS

- Languages: Mandarin (Native), English (Proficient), Cantonese (Beginner)
- Programming: Python, PyTorch, C, HTML, CSS
- Operating Systems: Linux, Windows, Mac OS
- Design: Adobe Photoshop/Lightroom/Premiere Pro
- Others: Prompt Engineering, Photography